EXTENSION 2

Stage 6

Calculus (Ext2), C1 Integration (Ext2)

Integration By Parts

Teacher: Nardene Dodd

Exam Equivalent Time: 55.5 minutes *(based on allocation of 1.5 minutes per mark)*

Questions

1. Calculus, EXT2 C1 2021 HSC 2 MC

Which expression is equal to $\int x^5 e^{7x} dx$?

A. $\dfrac{1}{7}x^5e^{7x} - \dfrac{5}{7}\int x^4e^{7x}dx$

B. $\dfrac{1}{7}\ x^5e^{7x} - \dfrac{5}{7}\int x^5e^{7x}dx$

C. $\dfrac{5}{7}x^4e^{7x} - \dfrac{5}{7}\int x^4e^{7x}dx$

D. $\dfrac{5}{7}\ x^4e^{7x} - \dfrac{5}{7}\int x^5e^{7x}dx$

2. Calculus, EXT2 C1 2015 HSC 6 MC

Which expression is equal to $\int x^2 \sin x\ dx$

A. $-x^2\cos x - \int 2x\cos x\ dx$

B. $-2x\cos x + \int x^2\cos x\ dx$

C. $-x^2\cos x + \int 2x\cos x\ dx$

D. $-2x\cos x - \int x^2\cos x\ dx$

3. Calculus, EXT2 C1 2019 HSC 3 MC

Which expression is equal to $\int x\cos x\ dx$?

A. $-x\sin x + \cos x + C$

B. $-x\sin x - \cos x + C$

C. $x\sin x + \cos x + C$

D. $x\sin x - \cos x + C$

4. Calculus, EXT2 C1 2011 HSC 1a

Find $\int x\ln x\ dx.$ ***(2 marks)***

5. Calculus, EXT2 C1 2003 HSC 1b

Use integration by parts to find $\int x^3\log_e x\ dx$ ***(3 marks)***

6. Calculus, EXT2 C1 2004 HSC 1a

Use integration by parts to find $\int xe^{3x}\ dx$. ***(2 marks)***

7. Calculus, EXT2 C1 2020 HSC 11b

Use integration by parts to evaluate $\int_1^e x\ln x\ dx$. ***(3 marks)***

8. Calculus, EXT2 C1 2006 HSC 1d

Evaluate $\int_0^2 te^{-t}\ dt.$ ***(3 marks)***

9. Calculus, EXT2 C1 2007 HSC 1c

Evaluate $\int_0^\pi x\cos x\ dx.$ ***(3 marks)***

10. Calculus, EXT2 C1 2009 HSC 1b

Find $\int xe^{2x}\ dx.$ ***(2 marks)***

11. Calculus, EXT2 C1 2014 HSC 11b

Evaluate $\int_0^{\frac{1}{2}} (3x-1)\ \cos\ (\pi x)\ dx$. *(3 marks)*

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12. Calculus, EXT2 C1 2016 HSC 11b

Find $\int xe^{-2x}\ dx$. *(3 marks)*

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13. Calculus, EXT2 C1 2017 HSC 12c

Find $\int x\tan^{-1}x\ dx$. *(3 marks)*

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14. Calculus, EXT2 C1 2016 HSC 12b

i. Differentiate $x\ f(x) - \int x\ f'(x)\ dx$. *(1 mark)*

ii. Hence, or otherwise, find $\int \tan^{-1}x\ dx$. *(2 marks)*

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15. Calculus, EXT2 C1 2005 HSC 1c

Use integration by parts to evaluate $\int_1^e x^7 \log_e x\ dx$. *(3 marks)*

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16. Calculus, EXT2 C1 2014 HSC 16c

Find $\int \frac{\ln x}{(1+\ln x)^2}\ dx$. *(3 marks)*

Worked Solutions

1. Calculus, EXT2 C1 2021 HSC 2 MC

$u = x^5 \qquad v\prime = e^{7x}$

$u\prime = 5x^4 \qquad v = \frac{1}{7}e^{7x}$

$$\int uv\prime = uv - \int u\prime v\ dx$$

$$= \frac{1}{7}\ x^5 e^{7x} - \frac{5}{7}\int x^4 e^{7x} dx$$

$\Rightarrow\ A$

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2. Calculus, EXT2 C1 2015 HSC 6 MC

$u = x^2, \qquad u\prime = 2x$

$v\prime = \sin x, \qquad v = -\cos x$

$$\int uv\prime\ dx = uv - \int u\prime v\ dx$$

$$= x^2(-\cos x) - \int 2x(-\cos x) dx$$

$$= -x^2 \cos x + \int 2x\cos x\ dx$$

$\Rightarrow\ C$

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3. Calculus, EXT2 C1 2019 HSC 3 MC

$u = x \qquad v\prime = \cos x$

$u\prime = 1 \qquad v = \sin x$

$$\int uv\prime\ dx = uv - \int u\prime v\ dx$$

$$= x\sin x - \int \sin x\ dx$$

$$= x\sin x + \cos x + C$$

$\Rightarrow\ C$

Worked Solutions

4. Calculus, EXT2 C1 2011 HSC 1a

Let $u = \ln x \quad v' = x$

$u' = \dfrac{1}{x} \quad v = \dfrac{x^2}{2}$

$$\begin{aligned}\int x\ln x\ dx &= \frac{x^2}{2}\cdot\ln x - \int \frac{x^2}{2}\times\frac{1}{x}\ dx\\ &= \frac{x^2\ln x}{2} - \frac{1}{2}\int x\ dx\\ &= \frac{x^2\ln x}{2} - \frac{x^2}{4} + c\end{aligned}$$

5. Calculus, EXT2 C1 2003 HSC 1b

$u = \log_e x, \quad u' = \dfrac{1}{x}$

$v' = x^3, \quad v = \dfrac{x^4}{4}$

$$\begin{aligned}\int x^3\log_e x\ dx &= uv - \int u'v\ dx\\ &= \frac{x^4}{4}\log_e x - \int \frac{1}{x}\cdot\frac{x^4}{4}\ dx\\ &= \frac{x^4\log_e x}{4} - \frac{1}{4}\int x^3 dx\\ &= \frac{x^4\log_e x}{4} - \frac{x^4}{16} + C\end{aligned}$$

6. Calculus, EXT2 C1 2004 HSC 1a

$u = x \quad u' = 1$

$v' = e^{3x} \quad v = \dfrac{1}{3}e^{3x}$

$$\begin{aligned}\int xe^{3x}\ dx &= uv - \int u'v\ dx\\ &= \frac{xe^{3x}}{3} - \frac{1}{3}\int e^{3x}\ dx\\ &= \frac{xe^{3x}}{3} - \frac{1}{9}e^{3x} + C\end{aligned}$$

7. Calculus, EXT2 C1 2020 HSC 11b

$u = \ln x \quad v' = x$

$u' = \dfrac{1}{x} \quad v = \dfrac{x^2}{2}$

$$\begin{aligned}\int_1^e x\ln x\ dx &= \left[\frac{x^2}{2}\cdot\ln x\right]_1^e - \int_1^e \frac{x^2}{2}\cdot\frac{1}{x}\ dx\\ &= \left[\frac{e^2}{2}\ln e - \frac{1}{2}\ln 1\right] - \int_1^e \frac{x}{2}\ dx\\ &= \frac{e^2}{2} - \left[\frac{x^2}{4}\right]_1^e\\ &= \frac{e^2}{2} - \left(\frac{e^2}{4} - \frac{1}{4}\right)\\ &= \frac{e^2+1}{4}\end{aligned}$$

8. Calculus, EXT2 C1 2006 HSC 1d

$u = t \quad u' = 1$

$v = -e^{-t}\ dt \quad v' = e^{-t}$

$$\begin{aligned}\int_0^2 te^{-t}\ dt &= \left[t\left(-e^{-t}\right)\right]_0^2 - \int_0^2 1\times\left(-e^{-t}\right)\ dt\\ &= \left[\left(-2e^{-2}\right) - 0\right] - \left[e^{-t}\right]_0^2\\ &= -\frac{2}{e^2} - \left(\frac{1}{e^2} - 1\right)\\ &= 1 - \frac{3}{e^2}\end{aligned}$$

9. Calculus, EXT2 C1 2007 HSC 1c

$u = x \quad u' = 1$

$v' = \cos x \quad v = \sin x$

$$\int uv'\ dx = uv - \int u'v\ dx$$

$$\begin{aligned}\therefore \int_0^\pi x\cos x\ dx &= [x\sin x]_0^\pi - \int_0^\pi 1\times\sin x\ dx\\ &= 0 + [\cos x]_0^\pi\\ &= -2\end{aligned}$$

10. Calculus, EXT2 C1 2009 HSC 1b

$$u = x \qquad u' = 1$$

$$v' = e^{2x} \qquad v = \frac{e^{2x}}{2}$$

$$\therefore \int xe^{2x}\ dx = x \cdot \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} \cdot 1\ dx$$

$$= \frac{xe^{2x}}{2} - \frac{e^{2x}}{4} + c$$

11. Calculus, EXT2 C1 2014 HSC 11b

Integrating by parts:

$$u = 3x - 1 \qquad u' = 3$$

$$v' = \cos(\pi x) \qquad v = \frac{1}{\pi}\sin(\pi x)$$

$$\int_0^{\frac{1}{2}} (3x-1)\ \cos\ (\pi x)\ dx$$

$$= \left[(3x-1)\frac{\sin\ (\pi x)}{\pi}\right]_0^{\frac{1}{2}} - \int_0^{\frac{1}{2}} 3 \times \frac{\sin\ (\pi x)}{\pi}\ dx$$

$$= \left(\frac{1}{2\pi}\ \sin\ \frac{\pi}{2} - 0\right) - \frac{3}{\pi}\left[\frac{-\cos\ (\pi x)}{\pi}\right]_0^{\frac{1}{2}}$$

$$= \frac{1}{2\pi} + \frac{3}{\pi^2}\left(\cos\ \frac{\pi}{2} - \cos\ 0\right)$$

$$= \frac{1}{2\pi} + \frac{3}{\pi^2}(0-1)$$

$$= \frac{1}{2\pi} - \frac{3}{\pi^2}$$

12. Calculus, EXT2 C1 2016 HSC 11b

Integrating by parts:

$$\text{Let} \quad u = x \qquad v' = e^{-2x}$$

$$u' = 1 \qquad v = -\frac{1}{2}e^{-2x}$$

$$\int xe^{-2x}dx = x \cdot -\frac{1}{2}e^{-2x} + \frac{1}{2}\int e^{-2x}dx$$

$$= -\frac{1}{2}xe^{-2x} - \frac{1}{4}e^{-2x} + c$$

13. Calculus, EXT2 C1 2017 HSC 12c

$$I = \int x\tan^{-1}x\ dx$$

$$\text{Let} \quad u = \tan^{-1}x \qquad v' = x$$

$$u' = \frac{1}{1+x^2} \qquad v = \frac{x^2}{2}$$

$$I = uv - \int u'v\ dx$$

$$= \tan^{-1}x \cdot \frac{x^2}{2} - \int \frac{1}{1+x^2} \cdot \frac{x^2}{2}\ dx$$

$$= \frac{x^2}{2}\ \tan^{-1}x - \frac{1}{2}\int \frac{x^2}{1+x^2}\ dx$$

$$= \frac{x^2}{2}\ \tan^{-1}x - \frac{1}{2}\int \frac{1+x^2-1}{1+x^2}\ dx$$

$$= \frac{x^2}{2}\ \tan^{-1}x - \frac{1}{2}\int 1 - \frac{1}{1+x^2}\ dx$$

$$= \frac{x^2}{2}\ \tan^{-1}x - \frac{1}{2}\left[x - \tan^{-1}x\right] + c$$

$$= \frac{x^2}{2}\ \tan^{-1}x - \frac{1}{2}x + \frac{1}{2}\tan^{-1}x + c$$

$$= \frac{1}{2}\left(x^2\tan^{-1}x - x + \tan^{-1}x\right) + c$$

14. Calculus, EXT2 C1 2016 HSC 12b

i. $$\frac{d}{dx}\left(x\ f(x) - \int x\ f'(x)\ dx\right)$$

$$= x\ f'(x) + f(x) - x\ f'(x)$$

$$= f(x)$$

ii. $$\int \tan^{-1}x\ dx = x\ \tan^{-1}x - \int \frac{x}{1+x^2}\ dx$$

$$= x\ \tan^{-1}x - \frac{1}{2}\ln\left(1+x^2\right) + c$$

15. Calculus, EXT2 C1 2005 HSC 1c

$$u = \log_e x, \qquad u' = \frac{1}{x}$$

$$v' = x^7, \qquad v = \frac{1}{8}x^8$$

$$\begin{aligned}
\int_1^e x^7 \log_e x \, dx &= uv - \int u'v \, dx \\
&= \left[\frac{x^8}{8} \cdot \log_e x\right]_1^e - \frac{1}{8}\int_1^e \frac{1}{x} \cdot x^8 dx \\
&= \left(\frac{e^8}{8}\log_e e - \frac{1}{8}\log_e 1\right) - \frac{1}{8}\left[\frac{1}{8}x^8\right]_1^e \\
&= \frac{e^8}{8} - \frac{1}{8}\left(\frac{e^8}{8} - \frac{1}{8}\right) \\
&= \frac{e^8}{8} - \left(\frac{e^8 - 1}{64}\right) \\
&= \frac{7e^8 + 1}{64}
\end{aligned}$$

16. Calculus, EXT2 C1 2014 HSC 16c

$$I = \int \frac{\ln x}{(1 + \ln x)^2} \, dx$$

Let $u = \ln x$

$$\frac{du}{dx} = \frac{1}{x} = \frac{1}{e^u}$$

$$du = \frac{1}{e^u} \, dx$$

$$dx = e^u \, du$$

$$\begin{aligned}
I &= \int \frac{ue^u}{(1+u)^2} \, du \\
&= \int \frac{(1+u)e^u}{(1+u)^2} \, du - \int \frac{e^u}{(1+u)^2} \, du \\
&= \int \frac{e^u}{1+u} \, du - \int \frac{e^u}{(1+u)^2} \, du
\end{aligned}$$

Using integration by parts:

$$u = -e^u \qquad u\prime = -e^u$$

$$v\prime = -(1+u)^{-2} \qquad v = (1+u)^{-1}$$

$$\begin{aligned}
\therefore I &= \int \frac{e^u}{1+u} \, du - \left(\frac{-e^u}{1+u} + \int \frac{e^u}{1+u} \, du\right) \\
&= \frac{e^u}{1+u} + c \\
&= \frac{x}{1 + \ln x} + c
\end{aligned}$$

♦♦ *Mean mark 31%.*
STRATEGY: This is as challenging as integration gets … a substitution followed by applying integration by parts. Note that simply substituting $u = \ln x$ gets you a full mark.